entanglement—and ask under what circumstances we can obtain emergent branches of the wave function that look like quantum fields propagating in a curved spacetime.

Up to this point in the book, basically everything we've talked about is either well understood and established doctrine (such as the essentials of quantum mechanics), or at least a plausible and respectable hypothesis (the Many-Worlds approach). Now we've reached the edge of what is safely understood, and will be venturing out into uncharted territory. We'll be looking at speculative ideas that might be important to understanding quantum spacetime and cosmology. But they might not be. Only years, possibly decades, of further investigation will reveal the answer with any confidence. By all means take these ideas as provocations to further thinking, and keep an eye on where the discussion goes in times to come, but keep in mind the intrinsic uncertainty that comes with wrestling with hard problems at the bleeding edge of our understanding.

Albert Einstein once mused to a colleague, "On quantum theory I use more brain grease than relativity." But it was his contributions to relativity that made him an intellectual superstar.

Like "quantum mechanics," "relativity" does not refer to a specific physical theory, but rather a framework within which theories can be constructed. Theories that are "relativistic" share a common picture of the nature of space and time, one in which the physical world is described by events happening in a single unified "spacetime." Even before relativity, it was still possible to talk about spacetime in Newtonian physics: there is three-dimensional space, and one dimension of time, and to locate an event in the universe you have to specify both where the event is in space and when it occurs in time. But before Einstein, there wasn't much motivation for combining them into a single four-dimensional concept. Once relativity came along, that became a natural step.

There are two big ideas that go under the name of "the theory of relativity," the special theory and the general theory. *Special relativity*, which came together in 1905, is based on the idea that everyone measures light to travel at the same speed in empty space. Combining that insight